

Ashutosh Tiwari

SOFTWARE ENGINEER | AI/ML | BACKEND

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PROFESSIONAL SUMMARY

Software Engineer with ~1.5 years shipping production software spanning backend services, AWS serverless automation, and applied AI/ML. Delivered a Google Ads integration (+30% conversions), automation removing ~260 tickets/year, and AI products (chatbot, voicebot, call analyzer) lifting international engagement 20%. M.Sc. in Mathematics & Computing; strong in Python, ML, and end-to-end delivery on Linux.

CORE SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, Bash, HTML/CSS

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, CrewAI, Deep Learning (CNN, RNN, Transformers), NLP, LLMs, Reinforcement Learning, Recommendation Systems, Predictive Modeling

Backend & APIs: REST APIs, Microservices, Webhooks, Flask, FastAPI, Django, Node.js

Cloud & DevOps: AWS (Lambda, Step Functions, EventBridge, ECS, Secrets Manager), Git/GitHub, CI/CD, Docker, Ubuntu/Linux, Jupyter, Colab

Data: Pandas, NumPy, ETL Pipelines, Data Warehousing, Web Scraping, Matplotlib, Seaborn, BigQuery

Databases: MySQL, PostgreSQL, MongoDB, BigQuery

PROFESSIONAL EXPERIENCE

Software Engineer, Simplotel Technologies

Mar 2025 – Present | Bangalore, India

- Designed, built, and shipped to production a Google Ads integration for Simplotel's hotel booking platform, automating sponsored-search campaigns for partner hotels and lifting booking conversions by 30%.
- Developed and deployed AI-driven customer-engagement products — a chatbot, a voicebot, and a call analyzer — raising engagement among international (out-of-India) clients by up to 20%.
- Engineered serverless automation across CMS and data-warehouse workflows using AWS Lambda, Step Functions, EventBridge, ECS, and Secrets Manager, eliminating ~260 recurring support tickets annually.
- Resolved a third-party VAPT (vulnerability assessment & penetration testing) audit of the hotel platform — remediating 5 critical, 12 high, and 15 medium application-security findings.

Data Analyst & ML Engineer, Vedasis

Sep 2022 – Nov 2022 | Ranchi, India

- Built scalable data pipelines ingesting social-media data through public APIs and custom web scrapers.
- Applied NLP techniques for text classification, clustering, and sentiment analysis on large unstructured datasets.
- Developed predictive ML and deep-learning models, including custom architectures for trend forecasting.

PROJECTS

- Payment Orchestrator** — Payment-orchestration service — multi-gateway routing, idempotency, an enforced state machine, HMAC signature verification, refunds, and signed webhooks; built in FastAPI with a live Cloudflare Workers edge twin. Python, FastAPI, SQLAlchemy, Razorpay. github.com/dsremo/pay-service
- Telemetry Anomaly Detection** — Satellite-telemetry anomaly-detection service — an LSTM/TCN ensemble with a per-user ops dashboard, alerting, and single sign-on; FastAPI backend, benchmarked on public ISS/ESA/SKAB datasets. Python, FastAPI, PyTorch. telemetry.dsremo.com
- ARIA** — A design study for a safety-first AI control architecture, framed for crewed and uncrewed spacecraft operations (reasoning + integration). Python, LLMs. github.com/dsremo/aria

EDUCATION

M.Sc. Mathematics & Computing, Birla Institute of Technology, Mesra

Jul 2019 – Jul 2024

Specialized in AI, Data Science & Machine Learning. Thesis: "Optimal Route Prediction using Deep Learning for Space Exploration."

CERTIFICATIONS & AWARDS

Certifications: Machine Learning Certification — Simplilearn; Deep Learning Specialization — DeepLearning.AI

Awards: Top 10 — Scaler Hackathon (2022); Paper Presentation — 3rd European Conference on MCDM (2022)